Check of your log $\frac{\vee}{k^+}$ terms (`Finite_terms_1_1.tex`) against BFKL

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1 Result

As written, it does not work.

- The symmetrized three-ρ terms do not cancel (Sec. 5).
- The two-ρ terms do not add up to your Eq. (LOBFKL). They do not add up to BFKL⊗LO−JIMWLK⊗LO either.
- Two signs of this that need no numerics:
  - a $\frac{1}{\epsilon}$ pole is left over (Sec. 4);
  - the kernel of the second line of (LOBFKL), $\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')K(x,y,z)$, appears in none of your terms (Sec. 6).

The reason. Your list contains only the *explicit* factors $\log\frac{\vee}{k^+}$, $\log\frac{k^+}{\vee}$, $\log\frac{\vee}{k^+ + \Lambda}$. In your rows $\log\frac{\vee}{k^+}$ also sits inside other logarithms that have $\vee$ in their argument, for example

$$\log\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{\Lambda(x-z)^2+k^+(x-w)^2}=\log\frac{\vee}{k^+}+\log\left[\left(1-\frac{k^+}{\vee}\right)\frac{(x-z)^2}{(x-w)^2}+\frac{k^+}{\vee}\right]+O\left(\frac{\Lambda}{k^+}\right),$$

and in $\log(\frac{\vee}{k^+}-1)$ and at the lower end $\xi\to\frac{k^+}{\vee}$ of ξ integrals. Row by row, against the exact coefficient of $\log\frac{\vee}{k^+}$ of the same row of `Finite_terms_1.tex` (Sec. 3):

- **5 rows are exactly right:** Group I Row II 2a, Row V parts 1, 2a, 2b UV, 2b finite.
- **6 rows are wrong:** Group I Rows III, IV; Group II Row I parts 1 and 2; Group III Rows I, II.
- **5 rows that do contain $\log\frac{\vee}{k^+}$ are marked NA:** Group II Row IV; Group III Row III (both parts), Rows V, VI.
- **Group I Row II part 1:** the coefficient is 1; it must be 2 ($\log\frac{\vee}{k^+}+\log(\frac{\vee}{k^+}-1)=2\log\frac{\vee}{k^+}+\log(1-\frac{k^+}{\vee})$).

With the exact coefficients. The complete list is in Sec. 7.

- The symmetrized three-ρ terms cancel.
- The two-ρ terms add up to

$$\boxed{\text{all } \log\frac{\vee}{k^+} \text{ terms} = \log\frac{\vee}{k^+}\Big(\mathcal{B}-\mathcal{J}\Big), \qquad \mathcal{B}=\text{BFKL}\otimes\text{LO (your Eq. (LOBFKL))}, \quad \mathcal{J}=\text{JIMWLK}\otimes\text{LO}.} \tag{1.1}$$

  This was verified exactly before (`LogVeeKplus_BFKL.pdf`: all 39 three-ρ structures cancel, all 37 two-ρ structures agree, pointwise in z). Here it is re-checked for the colour structure of Sec. 5, to 10^{-14}.
- It is $\mathcal{B}-\mathcal{J}$, not $\mathcal{B}$, because of your rule $\log\frac{k^+}{\Lambda}=\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}$. In every row the $\log\frac{\vee}{k^+}$ terms are $P-T$:
  - T is the row's $\log\frac{\vee}{\Lambda}$ coefficient (your evolution list, which sums to $\mathcal{J}$);
  - P is its large-p^+ limit (which sums to $\mathcal{B}$).
- BFKL multiplies $\log\frac{\vee}{k^+}$ once you split $\log\frac{\vee}{\Lambda}\,\mathcal{J}=\log\frac{k^+}{\Lambda}\,\mathcal{J}+\log\frac{\vee}{k^+}\mathcal{J}$:

$$\log\frac{\vee}{\Lambda}\,\mathcal{J}+\log\frac{\vee}{k^+}\big(\mathcal{B}-\mathcal{J}\big)=\log\frac{k^+}{\Lambda}\,\mathcal{J}+\log\frac{\vee}{k^+}\,\mathcal{B}. \tag{1.2}$$

- Your Eq. (LOBFKL) itself is correct: it is $\mathcal{B}$, your App. E, checked before to machine precision.

A correction to my previous notes (Sec. 8). The hidden $\log\frac{\vee}{k^+}$ at the lower end of the ξ integral of Group II Row IV was not in the table of `Finite_terms_simplified.pdf`, Sec. 6. As a result:

- its DGLAP term (5.5) still contains $\log\frac{\vee}{k^+}\,H_4$;
- the final results FS (11.1) and `Finite_terms_singlet.pdf` (7.1) counted $\frac{1}{2}\log\frac{\vee}{\Lambda}\,H_4$ twice.

Both notes are corrected and re-sent.

2 Notation and method

- $a=\frac{k^+}{\vee}$, $L_\vee=\log\frac{\vee}{k^+}$, $A=(x-z)^2$, $B=(x-w)^2$, $A'=(x'-z)^2$, $B'=(x'-w')^2$; $X'=\mathcal{K}(x'-w')$, $Y=\mathcal{K}(y-z)$, $X_w=\mathcal{K}(x-w)$, $X_z=\mathcal{K}(x-z)$, $Z_w=\mathcal{K}(z-w)$; $p_8=\frac{ig^4}{8\pi^5k^+}$, $p_{16}=\frac{ig^4}{16\pi^5k^+}$; $\mathcal{N}=\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5k^+}$.
- $\mathbb{T}^{ij}$, Σ_4, T_1, the brackets $[\dots]_{2b}$, $[\dots]_{2c}$, s, t_2, and the colour factors are those of `Finite_terms_simplified.pdf` (FS), Secs. 2 and 7.
- **Exact coefficient of $\log\frac{\vee}{k^+}$ of a row,** $C_{\rm row}=\lim\frac{\partial\,\text{row}}{\partial\log\vee}$ at fixed k^+, $\Lambda\to0$. It is computed in three independent ways:
  - the closed forms of FS Sec. 6, validated there: row $-\,C_{\rm row}L_\vee$ has a limit for $\frac{\vee}{k^+}=10^4\dots10^8$;
  - here, by differentiating your rows of `Finite_terms_1.tex` numerically at $\frac{\vee}{k^+}=10^7,10^9$ (agreement in the last column of Table 1);
  - the rows themselves were checked earlier against the exact p^+ integrals of the first-principles integrands (`../finite/verify.py`).
- Your terms are transcribed term by term in `user_lvk.py` and evaluated at random transverse points, with random classical charges and random adjoint $U=U^T$ (your convention). The normalization is that of the file: $\frac{1}{(2\pi)^3}$, prefactor, phase, colour, no c.c.

3 Row by row

Table 1: Your term vs. the exact coefficient of $\log\frac{\vee}{k^+}$ of the same row, $|\text{yours}-\text{exact}|/|\text{exact}|$ at three random configurations.

row	yours vs. exact	verdict	exact vs. $\partial_{\log\vee}$(file)
G1 Row II part 1	coefficient 1 instead of 2 (analytic)	**wrong**	—
G1 Row II 2a	0, 0, 0	**correct**	$\leq 2.6\times10^{-15}$
G1 Row III	8, 0.047, 1.5	**wrong**	$\leq 8.2\times10^{-8}$
G1 Row IV	1.8, 0.088, 11	**wrong**	$\leq 2.7\times10^{-7}$
G1 Row V part 1	0, 0, 0	**correct**	$\leq 1.0\times10^{-15}$
G1 Row V 2a	3.1×10^{-16}, 2.0×10^{-16}, 1.9×10^{-16}	**correct**	$\leq 4.1\times10^{-16}$
G1 Row V 2b UV	same as the file (analytic)	**correct**	—
G1 Row V 2b finite	0, 2.6×10^{-16}, 2.0×10^{-17}	**correct**	$\leq 1.0\times10^{-15}$
G2 Row I part 1	1.5, 0.053, 4.6	**wrong**	$\leq 1.8\times10^{-7}$
G2 Row I part 2	0.9, 0.9, 1.1e+02	**wrong**	$\leq 3.0\times10^{-6}$
G2 Row IV	NA; the file has $-\frac{\alpha_s}{2\pi}2N_cL_\vee\mathrm{LO}[W]+L_\vee H_4$	**missing**	—
G3 Row I	1.8, 0.088, 11	**wrong**	$\leq 2.7\times10^{-7}$
G3 Row II	1.8, 0.088, 11	**wrong**	$\leq 2.7\times10^{-7}$
G3 Row III part 1	NA (yours = 0)	**missing**	$\leq 3.6\times10^{-1}$
G3 Row III part 2	NA (yours = 0)	**missing**	$\leq 6.6\times10^{-2}$
G3 Row V	NA (yours = 0)	**missing**	$\leq 3.6\times10^{-1}$
G3 Row VI	NA (yours = 0)	**missing**	$\leq 3.6\times10^{-1}$

For Group III Rows III, V, VI the last column carries the bounded oscillating terms $e^{-i\kappa(1-a)/a}$, which vanish only after $\int_{y'}$. FS Sec. 6 removes them explicitly (spread $\leq 2\times10^{-4}$).

What is missing in each row. The exact coefficients, with what you have marked:

$$\text{G1 Row II part 1:}\quad C = 2\times\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4k^+}\int e^{-ik\cdot(w'-w)}\,X'\cdot X_w\Big[-\frac{2}{\epsilon}-2\gamma-\log\frac{B\mu^2}{4}\Big]$$
$$\times\,\mathcal{C}_L^a\big[U^{ab}(w)-U^{ab}(x)\big]\rho^b(x)\qquad\text{(yours: }1\times\text{)},\tag{3.1}$$

$$\text{G1 Row IV:}\quad C = p_8\big[X'^iY^j\mathbb{T}^{ij}-\tfrac{X'\cdot Y}{2(z-w)^2}\big]$$
$$= p_8\big[(X'\cdot Z_w)(Y\cdot X_z)-(X'\cdot X_w)(Y\cdot X_z)-(X'\cdot X_w)(Y\cdot Z_w)\big],$$
$$\text{yours: } p_8\big[-\tfrac{X'\cdot Y}{2(z-w)^2}\big];$$
$$\text{missing: } p_8X'^iY^j\mathbb{T}^{ij}\ (\text{inside }\log\tfrac{(\vee-k^+)A+k^+B}{\Lambda A+k^+B}),\tag{3.2}$$

$$\text{G3 Rows I, II:}\quad C = -p_8\big[X'^iY^j\mathbb{T}^{ij}-\tfrac{X'\cdot Y}{2(z-w)^2}\big]\ (\text{same log}),\tag{3.3}$$

$$\text{G2 Row I part 1:}\quad C = -p_{16}\big[X'^iY^j\mathbb{T}^{ij}-\tfrac{X'\cdot Y}{2(z-w)^2}\big]_w\mathcal{O}_A$$
$$+\,p_{16}\big[\dots\big]_{w\leftrightarrow w'}\mathcal{O}_B\ (\text{same log}),\tag{3.4}$$

$$\text{G1 Row III:}\quad C = p_8\Big[(X'\cdot X_w)(Y\cdot Z_w-Y\cdot X_z)+\frac{X'\cdot Y}{(z-w)^2}-\Sigma_4\Big],$$
$$\text{yours:}\quad -p_{16}T_1+p_8(X'\cdot X_w)(Y\cdot Z_w)+p_8\frac{X'\cdot Y}{(z-w)^2}$$
$$+\,p_8(X'\cdot Y)\frac{[(z-w)\cdot(x-w)]A-[(z-w)\cdot(x-z)]B}{(z-w)^2(B-A)^2},$$
$$\text{missing, from }\log\big|\tfrac{\vee A-k^+B}{(\Lambda+k^+)A-k^+B}\big|,\ \log\big|\tfrac{k^+B-\vee A}{k^+B-\Lambda A}\big|,$$
$$\log\big|\tfrac{Bk^+-A\vee}{Bk^+-A(k^++\Lambda)}\big|:$$
$$+\,p_{16}T_1-p_8(X'\cdot X_w)(Y\cdot X_z)-p_8\Sigma_4$$
$$-\,p_8(X'\cdot Y)\frac{[(z-w)\cdot(x-w)]A-[(z-w)\cdot(x-z)]B}{(z-w)^2(B-A)^2},\tag{3.5}$$

$$\text{G2 Row I part 2:}\quad C = \frac{g^4}{8\pi^5k^+}\Big[-\frac{[\dots]_{2b}}{BB'}+\frac{[\dots]_{2c}}{AA'}\Big]\mathcal{O}_{58}.\tag{3.6}$$

- **G2 Row I part 2, your 2d, 2e, 2f:** their explicit $\log\frac{\vee}{k^++\Lambda}$ cancel exactly against the $\log\frac{\vee}{k^+}$ inside

$\log\frac{(\vee-k^+)A+k^+B}{\Lambda A+k^+B}$ and its primed partner in the same lines, so 2d, 2e, 2f give 0. The coefficient comes from 2b and 2c, whose logarithms you did not count.

- **Rows marked NA:**

$$\text{G3 Rows III(1), V, VI:}\quad C=\frac{1}{(2\pi)^3}\frac{c_C}{k^+}\int e^{-ik\cdot(y'-w)}\,t_2\,\mathcal{O}_C\quad(\text{from } E_1(i\kappa\tfrac{1-a}{a})),$$
$$c_{\text{III(1)}}=\tfrac{ig^4}{16\pi^5},\ c_{\text{V}}=c_{\text{VI}}=\tfrac{ig^4}{8\pi^5},\tag{3.7}$$
$$\text{G3 Row III part 2:}\quad C=\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\int e^{-ik\cdot(y'-w)}\Big(-\frac{s}{k^+B}\Big)\mathcal{O}_{\text{III(2)}},\tag{3.8}$$
$$\text{G2 Row IV:}\quad C=-\frac{\alpha_s}{2\pi}2N_c\,\text{LO}[W]+H_4\qquad(\text{Sec. 8}).\tag{3.9}$$

4 The poles

The $\frac{1}{\epsilon}$ poles multiplying $\log\frac{\vee}{k^+}$ come from three rows. In units of LO, with c.c. ($\hat{E}_r=\frac{2}{\epsilon}+\ldots$; FS Eqs. (4.1), (4.2), (5.1)):

row	your list	exact
G1 Row II part 1	$\frac{\alpha_s N_c}{4\pi}L_\vee\,\text{LO}[\hat{E}_X+\hat{E}_{X'}]$	$\frac{\alpha_s N_c}{2\pi}L_\vee\,\text{LO}[\hat{E}_X+\hat{E}_{X'}]$
G1 Row V 2b UV	$-\frac{\alpha_s N_c}{\pi}L_\vee\,\text{LO}[\hat{E}_X+\hat{E}_{X'}]$	the same
G2 Row IV	NA	$+\frac{\alpha_s N_c}{\pi}L_\vee\,\text{LO}[\hat{E}_\Delta]$
pole	$-\frac{3\alpha_s N_c}{\pi}\frac{1}{\epsilon}L_\vee\,\text{LO}$	0; finite part $\frac{\alpha_s N_c}{2\pi}L_\vee\,\text{LO}\big[\log\frac{(y-y')^4}{(x-y)^2(x'-y')^2}\big]$

Your list leaves a pole; BFKL⊗LO has none.

5 Three ρ's

The symmetrized three-ρ colour of your rows is that of FS Secs. 8, 9. One of its structures does not depend on z:

$$C_w\equiv f^{abc}\,\mathcal{C}^a_L\,\big(U(w)\rho(x)\big)^b\big(U(w)\rho(y)\big)^c,\qquad \mathcal{C}^a_L=\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x').\tag{5.1}$$

- It comes from two places (checked numerically to 10^{-15}):
  - Group I Row II 2a, whose symmetric colour contains $2C_w$;
  - the $\mathcal{K}_{\text{IV}}$ family (G1 IV, G3 I, II, G2 I(1) block A), whose combined symmetric colour is $-C_w+f^{acd}\mathcal{C}^a_L(U(z)\rho(y))^c(U(x)\rho(x))^d$ (FS Eq. (8.2)).
- Since C_w does not depend on z, its total coefficient is the z integral of the kernels, antisymmetrized in $x\leftrightarrow y$. It must vanish:

$$-\frac{ig^4}{32\pi^4k^+}\,2\,X'\cdot\mathfrak{F}(w;x,y)\;-\;\frac{ig^4}{8\pi^5k^+}\int_z\big[\text{kernel}\big]_{x\leftrightarrow y\ \text{antisym.}}=0,\tag{5.2}$$
$$\mathfrak{F}^i(w;x,y)=\mathcal{K}^i(x-w)\log\frac{(w-y)^2}{(x-y)^2}-\mathcal{K}^i(y-w)\log\frac{(w-x)^2}{(x-y)^2}\quad(\text{your Row II 2a}).$$

- **Exact kernel** $X'^iY^j\mathbb{T}^{ij}-\frac{X'\cdot Y}{2(z-w)^2}$: numerically, $\int_z[\ldots]_A=-\frac{\pi}{2}X'\cdot\mathfrak{F}$ to $\le 2.4\times10^{-14}$. The structure cancels.
- **Your kernel** $-\frac{X'\cdot Y}{2(z-w)^2}$: the integral diverges at $z\to w$. With a cut $|z-w|>\varepsilon$ it grows like $\log\frac{1}{\varepsilon}$ (6.40, 10.45, 14.49; -0.20, -0.32, -0.45; -7.78, -11.91, -16.03 for $\varepsilon=10^{-2},10^{-3},10^{-4}$). Nothing else in your list carries C_w, so your three-ρ terms cannot cancel.

6 Two ρ's

- The second line of your Eq. (LOBFKL) has the kernel $\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')\,K(x,y,z)$: the measured gluon is emitted by the soft gluon in the amplitude *and* in the conjugate amplitude.
- In the exact list it comes from Group II Row I part 2, through the first term of $[\ldots]_{2c}/(AA')$, $\frac{(x'-z)\cdot(x-z)}{AA'}\frac{(z-w)\cdot(z-w')}{(z-w)^2(z-w')^2}$, and from H_4 of Group II Row IV.
- No term of your list has the factor $(z-w)\cdot(z-w')$: 2c is not in your list and Row IV is NA. So your two-ρ terms cannot produce this line.

7 The complete list and what it gives

Complete list. Your list, with:

1. Group I Row II part 1: coefficient 2;
2. Group I Rows III, IV; Group II Row I parts 1, 2; Group III Rows I, II: the exact C of Sec. 3 in place of yours;
3. added: Group II Row IV ($-\frac{\alpha_s}{2\pi}2N_cL_\vee\mathrm{LO}[W]+L_\vee H_4$), Group III Rows III(1), III(2), V, VI.

These are exactly the L_* terms of FS Secs. 6, 7 plus H_4.

What it gives. Eq. (1.1): $L_\vee(\mathcal{B}-\mathcal{J})$.

- Verified exactly in `LogVeeKplus_BFKL.pdf`:
  - symmetrized three-ρ: 39 structures, all zero;
  - two-ρ: 37 kernel structures, all equal to $\mathcal{B}-\mathcal{J}$;
  - the same at two more random configurations (largest difference 8×10^{-14}).
- There the rows were written pointwise in z from your notes; here the rows of `Finite_terms_1.tex` are the same rows after the p^+ integration.
- Sec. 5 re-checks one structure directly in your finite-file form.

Why not $\mathcal{B}$ alone. For every row,

$$\int_\Lambda^\vee \frac{dp^+}{p^+}G\Big(\frac{p^+}{k^+}\Big)=G_T\log\frac{k^+}{\Lambda}+G_P\log\frac{\vee}{k^+}+\dots=G_T\log\frac{\vee}{\Lambda}+(G_P-G_T)\log\frac{\vee}{k^+}+\dots \tag{7.1}$$

- Your evolution list is $\sum G_T=\mathcal{J}$.
- Your $\log\frac{\vee}{k^+}$ list is $\sum(G_P-G_T)=\mathcal{B}-\mathcal{J}$.
- In total: $\log\frac{k^+}{\Lambda}\mathcal{J}+\log\frac{\vee}{k^+}\mathcal{B}$. At mid rapidity this is $\frac12\log\frac{\vee}{\Lambda}(\mathcal{J}+\mathcal{B})$, as in your corrected evolution file.

8 Correction to my previous notes: the lower end of Group II Row IV

The term. The ξ integral of Group II Row IV (FS Eq. (5.1)) runs from $\frac{k^+}{\vee}$.

- At small ξ (soft gluon much faster than the measured one), LO_ξ probes the transverse momentum $\frac{k}{\xi}$.
- The only power-law tail comes from $\log(y'-y)^2$ in ℓ_Δ:

$$\int d^2\Delta\, e^{-i\mathbf{q}\cdot\Delta}\log(\Delta^2)\,h(\Delta)\ \xrightarrow{q\to\infty}\ -\frac{4\pi}{q^2}h(0) \tag{8.1}$$

  (checked: $q^2\times$integral$/(-4\pi h(0))$ = 1.0209, 1.0022, 1.0002 (Gaussian h), 1.1119, 1.0091, 1.0008 ($h=(1+\Delta^2)^{-2}$) at $q=10,30,100$).
- So $\mathrm{LO}_\xi[\ell_\Delta]\propto\xi^2$, and with $P_{gg}^{(r)}\to\frac{2N_c}{\xi}$ the integral $\int_{k^+/\vee}\frac{d\xi}{\xi^2}P\,\mathrm{LO}_\xi$ gives $\log\frac{\vee}{k^+}$.

Exactly:

$$-\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}P_{gg,+}^{(r)}(\xi)\,\mathrm{LO}_\xi[\ell_\Delta]=L_\vee\,H_4+(\text{finite for }\tfrac{k^+}{\vee}\to0), \tag{8.2}$$

$$\boxed{\begin{aligned}H_4=2N_c\mathcal{N}\int_{x,x',z,w,w'}&e^{-ik\cdot(w'-w)}\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\\&\times\big[U^{e'c}(z)-U^{e'c}(x')\big]\big[U^{ec}(z)-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x)\end{aligned}} \tag{8.3}$$

- This is the region-P term "II.IV P" of `LogVeeKplus_BFKL.pdf` (the normalizations agree: both are $\frac{2N_cg^4}{4\pi^3k^+k^2}\times\frac{1}{(2\pi)^3}\int_{x,x',z}\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,\mathcal{O}|_{y=y'=z}$).
- In the singlet projection: $H_4=2N_c\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K(w,w',z)K(x,x',z)\,\mathcal{P}(x,x')\,\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(x'))(U(z)-U(x))\}$.

The correction.

- In FS (11.1): the DGLAP term is replaced by $\big[-\frac{\alpha_s}{2\pi}\int_a^1 \cdots - L_\vee H_4\big] + L_* H_4$, i.e. $-\frac{1}{2}\log\frac{\vee}{\Lambda}\, H_4$ is added.
- In `Finite_terms_singlet.pdf` (7.1), where all L_* terms are collected in the remainder: the DGLAP term is replaced by $\big[-\frac{\alpha_s}{2\pi}\int_a^1 \cdots - L_\vee H_4\big]$.
- With H_4 included, $\sum_{\text{rows}} C_{\text{row}} = \mathcal{N}(\mathcal{B} - \mathcal{J})$ holds term by term, as FS (6.2) states.

9 Files

In `nlo_gluon_paper/bfkl_check/`:

- `user_lvk.py`: your list, term by term;
- `rowcheck.py`: Table 1;
- `cw_test.py`: Sec. 5;
- `ft_asym.py`: Sec. 8;
- `make_note.py`: this note.

Your file as received is `input/Finite_terms_1_1.tex`.